

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Publication

- “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023** (3rd International Conference on Intelligent Technologies), applied deep transfer learning on neuroimaging data for medical image classification (first author).

Skills

Languages

Python, JavaScript, TypeScript, C++, SQL, Bash

Research Methods

Experimental design, controls & ablations, statistical analysis (bootstrap CIs, McNemar’s test), data cleaning, research documentation, technical writing

Data & Analysis

Python, pandas, NumPy, matplotlib (visualization), statistical analysis, Apache Spark, scikit-learn

AI & Agents

AI agents / assistants, LLM agents, agentic workflows, evaluation methodology, RAG, OpenAI / Groq APIs, PyTorch

Web & Tools

React, JavaScript (web-based systems), Git, Jupyter, LaTeX, FastAPI

Research & Projects

Refusal Decay - Empirical Study of AI Behavior

UMass CS 602 Research Project (graded full marks)

Python, PyTorch, statistical analysis | controlled experiment + behavioral measurement

- Designed and ran a controlled **empirical study** of how an AI model’s behavior shifts under attack: defined conditions, multi-seed random/orthogonal **controls**, and a benign positive control, then measured behavioral outcomes across hundreds of trials.
- Conducted **statistical analysis** (bootstrap 95% CIs, McNemar’s test) and validated heuristic labels against an independent classifier (**87.5% agreement**), documenting threats to validity and findings across a structured multi-report research arc.

AI Assistants & Agents - AgenticSearch, cf-scout

ciir_agentic_search, cf_ai_research_scout

Python, FastAPI, TypeScript, LLMs | AI agents that assist users

- Built **AgenticSearch**, an AI agent that turns open queries into verified, provenance-backed results so a person can trust and audit what the AI produced, hardened with **150+ automated tests**.
- Built **cf-scout**, an AI research assistant that works alongside the user, fetching sources and streaming multi-step research digests in real time, giving direct exposure to how people interact with and rely on an AI teammate.

AutoEval - Reproducible Evaluation & Reporting

github.com/Deva-1903/AutoEval

Python, evaluation methodology, Git automation

- Built an agent that scores systems on a holdout-heavy objective and commits **auditable report cards** back to Git for a reproducible analysis workflow.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) to keep the evaluation trustworthy and the results well-documented.

KG2RAG-Enhanced - Quantitative Evaluation (team)

github.com/Deva-1903/KG2RAG-685-NLP

Python, statistical evaluation | UMass CS 685 Advanced NLP

- Led the retrieval component of a team research project and ran a **4,905-question evaluation** with **95% binomial confidence intervals**, reporting a **+2.68%** improvement, quantitative analysis of system behavior in Python.

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models; replaced spreadsheet workflows and accelerated iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.
- Worked **cross-functionally** with clinical, product, and AI teams to scope and ship features, and ran **knowledge-transfer** sessions that onboarded a new developer, translating needs across non-engineering and engineering stakeholders.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.